

Research Paper-2

Visual Instruction Tuning (LLaVA)Liu et al., 2023

Summary: This paper is about LLaVA (Large Language and Visual Assistant) this is an AI Model that can understand the image and the text both at the same time and give answer for each one of them.

Before LLaVA, many AI Models could either understand text only (like ChatGPT) or understand images only. There were very few models that could understand questions/text related to an image.

Therefore, the authors used GPT-4 to create a large dataset of image-related questions and answers.

They combined—CLIP for understanding images and used Vicuna/LLaMA (Large language model Meta AI) for understanding language. And then they trained the model to answer the image-based questions, describe images and perform reasoning.

How does it Work?

1. User gives the image and question related to the image.
for example there is an image of a dog playing with a ball on the grass and the question asked is what is the dog doing? So LLaVA will answer that the dog is playing with a ball on the grass.
2. Now, CLIP extracts image features.
3. Vicuna processes the question.
4. LLaVA combines both and generates an answer.

- **Model Used:** LLaVA (Large language and Visual Assistant)
- Vision Encoder used: CLIP-ViT-L/14
- Language Model Component: Vicuna
- Vision-language connector type: Simple linear layer with trainable projection matrix.
- Connection/integration: Visual features integrated into language model's word embedding space.
- Architectural modifications/improvements: None mentioned
- Unified visual representation: Not mentioned

Accuracy: LLaVA achieved 90.92% accuracy on the ScienceQA (large multimodal question-answering dataset used to test how well AI models can solve science questions. It contains questions from elementary and middle/high school science subjects and may include text, images, diagrams, or both.) dataset and reached 92.53% accuracy when combined with GPT-4, outperforming previous state-of-the-art methods.

Dataset Used: LLaVA was pre-trained on the CC-595K dataset, instruction-tuned on the GPT-4-generated LLaVA-Instruct-158K dataset, and evaluated on the ScienceQA benchmark.

- Data generation: Synthetic using ChatGPT/GPT-4
- Training procedure: Pre-training on CC-595K for 1 epoch, fine-tuning on LLaVA-Instruct-158K for 3 epochs
- Learning rate: Pre-training $2e-3$, fine-tuning $2e-5$
- Batch size: Pre-training 128, fine-tuning 32

Strengths:

- Supports image + text understanding
- Strong visual reasoning and instruction following
- High ScienceQA accuracy (90.92%)
- Large GPT-4-generated training dataset
- Open-source and easy for research use
- Performs better than many earlier multimodal models

Limitations:

- Hallucinates and may generate incorrect information.
- Struggles to recognize small objects and fine image details.
- Sometimes fails to understand the complete context of an image.
- Inherits biases from pretrained vision and language models.
- Depends heavily on GPT-4 generated training data.
- Cannot access real-time or updated external knowledge.

Research Gap

- Need for better fine-grained image understanding.
- Need to reduce hallucinations and improve answer reliability.
- Need stronger multimodal reasoning capabilities.
- Need integration with real-time external knowledge sources.
- Need support for high-resolution image analysis.
- Need methods to reduce bias and improve fairness.

Future Scope: Future work can focus on reducing hallucinations, improving fine-grained visual understanding, integrating real-time knowledge retrieval, supporting high-resolution images and videos, and enhancing multimodal reasoning capabilities.

Comparison Table between our Topic and Paper-2

Aspect	LLaVA (2023)	Your Proposed Research	
Goal	General image understanding and visual chatbot	Crop disease classification in agriculture	
Domain	General-purpose images (COCO, ScienceQA)	Agricultural leaf disease images	
Focus	Building a multimodal AI assistant	Benchmarking VLMs for practical agricultural use	
Dataset	CC3M, COCO, ScienceQA	PlantVillage + PlantDoc	
Models	Mainly LLaVA architecture	Multiple VLMs (LLaVA, CLIP, InternVL2, etc.) vs CNNs	
Evaluation	Visual reasoning and question answering	Disease classification accuracy and real-world deployment	
Application	General AI assistant	Helping farmers identify crop diseases	

How our Topic is Better

- Focuses on a real-world agricultural problem instead of general image understanding.
- Compares multiple VLMs, not just LLaVA.
- Tests whether VLMs are practical for crop disease detection.
- Includes real-world field images (PlantDoc), making results more realistic.
- Provides guidance for agricultural researchers and farmers.
- Fully open-source and reproducible using free models and datasets.

What Improvements our Research Makes

- Evaluates VLMs specifically in agriculture, which LLaVA does not.
- Compares zero-shot VLMs with strong supervised CNN baselines.
- Analyzes failure cases of VLMs on different disease classes.
- Studies model performance on both clean and real-world datasets.
- Investigates whether VLMs can reduce the need for large labeled agricultural datasets.

What LLaVA Does Better Than our Topic

- Can hold conversations and answer image-based questions.
- Performs visual reasoning and detailed image descriptions.
- More general-purpose and applicable across many domains.
- Supports multimodal chat rather than only classification tasks.